

11.29. A.C. Through Pure Ohmic Resistance Alone

The circuit is shown in Fig. 11.56. Let the applied voltage be given by the equation.

$$v = V_m \sin \theta = V_m \sin \omega t \quad \dots(i)$$

Let R = ohmic resistance ; i = instantaneous current

Obviously, the applied voltage has to supply ohmic voltage drop only. Hence for equilibrium

$$v = iR;$$

Putting the value of 'v' from above, we get $V_m \sin \omega t = iR$; $i = \frac{V_m}{R} \sin \omega t \quad \dots(ii)$

Current 'i' is maximum when $\sin \omega t$ is unity $\therefore I_m = V_m/R$ Hence, equation (ii) becomes, $i = I_m \sin \omega t \quad \dots(iii)$

Comparing (i) and (ii), we find that the alternating voltage and current are in phase with each other as shown in Fig. 11.57. It is also shown vectorially by vectors V_R and I in Fig. 11.54.

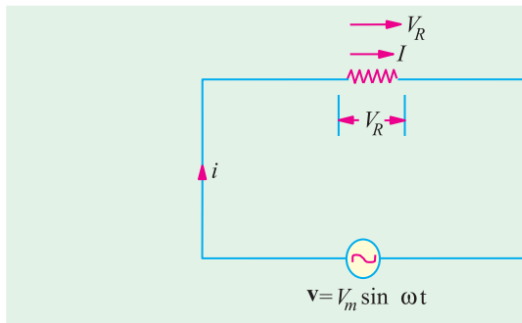


Fig. 11.56

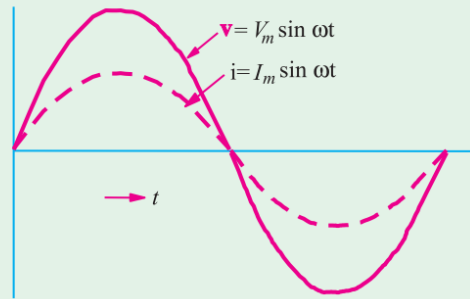


Fig. 11.57

11.30. A.C. Through Pure Inductance Alone

Whenever an alternating voltage is applied to a purely inductive coil*, a back e.m.f. is produced due to the self-inductance of the coil. The back e.m.f., at every step, opposes the rise or fall of current through the coil. As there is no ohmic voltage drop, the applied voltage has to overcome this self-induced e.m.f. only. So at every step

$$v = L \frac{di}{dt}$$

Now $v = V_m \sin \omega t$

$$\therefore V_m \sin \omega t = L \frac{di}{dt} \therefore di = \frac{V_m}{L} \sin \omega t dt$$

Integrating both sides, we get $i = \frac{V_m}{L} \int \sin \omega t dt$

$$= \frac{V_m}{\omega L} (\cos \omega t)$$

...(constant of integration = 0)

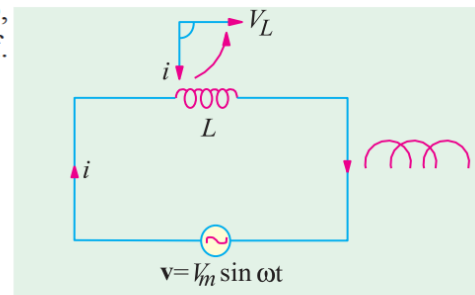


Fig. 11.59

$$\therefore = \frac{V_m}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right) = \frac{V_m}{X_L} \sin (\omega t - \pi/2) \quad \dots(ii)$$

Max. value of i is $I_m = \frac{V_m}{\omega L}$ when $\sin \left(\omega t - \frac{\pi}{2} \right)$ is unity.

Hence, the equation of the current becomes $i = I_m \sin (\omega t - \pi/2)$.

So, we find that if applied voltage is represented by $v = V_m \sin \omega t$, then current flowing in a *purely* inductive circuit is given by $i = I_m \sin \left(\omega t - \frac{\pi}{2} \right)$

11.32. A.C. Through Pure Capacitance Alone

When an alternating voltage is applied to the plates of a capacitor, the capacitor is charged first in one direction and then in the opposite direction. When reference to Fig. 11.62, let

v = p.d. developed between plates at any instant

q = Charge on plates at that instant.

Then

$$q = Cv$$

...where C is the capacitance

$$= C V_m \sin \omega t$$

...putting the value of v .

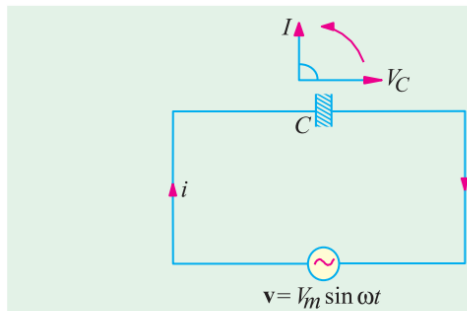


Fig. 11.62

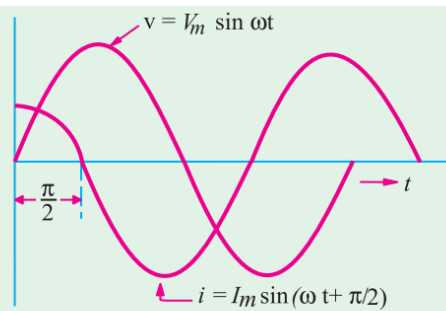


Fig. 11.63

Now, current i is given by the rate of flow of charge.

$$\therefore i = \frac{dq}{dt} = \frac{d}{dt} (C V_m \sin \omega t) = C V_m \cos \omega t \text{ or } i = \frac{V_m}{1/\omega C} \cos \omega t = \frac{V_m}{1/\omega C} \sin \left(\omega t + \frac{\pi}{2} \right)$$

Obviously, $I_m = \frac{V_m}{1/\omega C} = \frac{V_m}{X_C}$

$$i = I_m \sin \left(\omega t + \frac{\pi}{2} \right)$$

The denominator $X_C = 1/\omega C$ is known as *capacitive reactance* and is in ohms if C is in farad and ω in radian/second. It is seen that if the applied voltage is given by $v = V_m \sin \omega t$, then the current is given by $i = I_m \sin (\omega t + \pi/2)$.

13.9. Resistance, Inductance and Capacitance in Series

The three are shown in Fig. 13.33 (a) joined in series across an a.c. supply of r.m.s. voltage V .

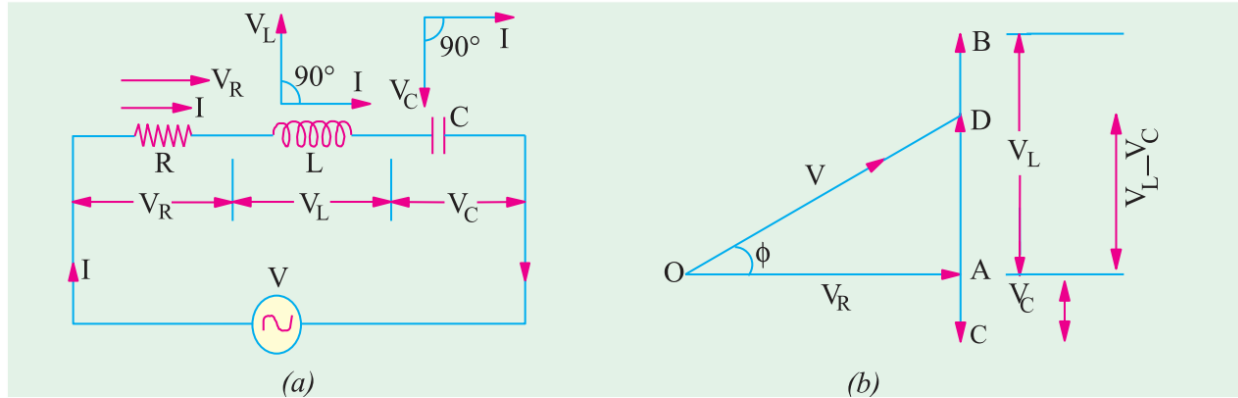


Fig. 13.33

Let $V_R = IR =$ voltage drop across R —in phase with I
 $V_L = I.X_L =$ voltage drop across L —leading I by $\pi/2$
 $V_C = I.X_C =$ voltage drop across C —lagging I by $\pi/2$

In voltage triangle of Fig. 13.33 (b), OA represents V_R , AB and AC represent the inductive and capacitive drops respectively. It will be seen that V_L and V_C are 180° out of phase with each other *i.e.* they are in direct opposition to each other.

Subtracting $BD (= AC)$ from AB , we get the net reactive drop $AD = I(X_L - X_C)$

The applied voltage V is represented by OD and is the vector sum of OA and AD

$$\therefore OD = \sqrt{OA^2 + AD^2} \text{ or } V = \sqrt{(IR)^2 + (IX_L - IX_C)^2} = I \sqrt{R^2 + (X_L - X_C)^2}$$

$$\text{or } I = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{V}{\sqrt{R^2 + X^2}} = \frac{V}{Z}$$

The term $\sqrt{R^2 + (X_L - X_C)^2}$ is known as the impedance of the circuit. Obviously,

$$(\text{impedance})^2 = (\text{resistance})^2 + (\text{net reactance})^2$$

$$\text{or } Z^2 = R^2 + (X_L - X_C)^2 = R^2 + X^2$$

13.10. Resonance in R-L-C Circuits

We have seen from Art. 13.9 that net reactance in an R - L - C circuit of Fig. 13.40 (a) is

$$X = X_L - X_C \text{ and } Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + X^2}$$

Let such a circuit be connected across an a.c. source of constant voltage V but of frequency varying from zero to infinity. There would be a certain frequency of the applied voltage which would make X_L equal to X_C in magnitude. In that case, $X=0$ and $Z=R$ as shown in Fig. 13.40 (c). Under this condition, the circuit is said to be in electrical resonance.

Calculation of Resonant Frequency

The frequency at which the net reactance of the series circuit is zero is called the resonant frequency f_0 . Its value can be found as under : $X_L - X_C = 0$ or $X_L = X_C$ or $\omega_0 L = 1/\omega_0 C$

$$\text{or} \quad \omega_0^2 = \frac{1}{LC} \quad \text{or} \quad (2\pi f_0)^2 = \frac{1}{LC} \quad \text{or} \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

If L is in henry and C in farad, then f_0 is given in Hz .

When a series R - L - C circuit is in resonance, it possesses minimum impedance $Z = R$. Hence, circuit current is maximum, it being limited by value of R alone. The current $I_0 = V/R$ and is in phase with V .

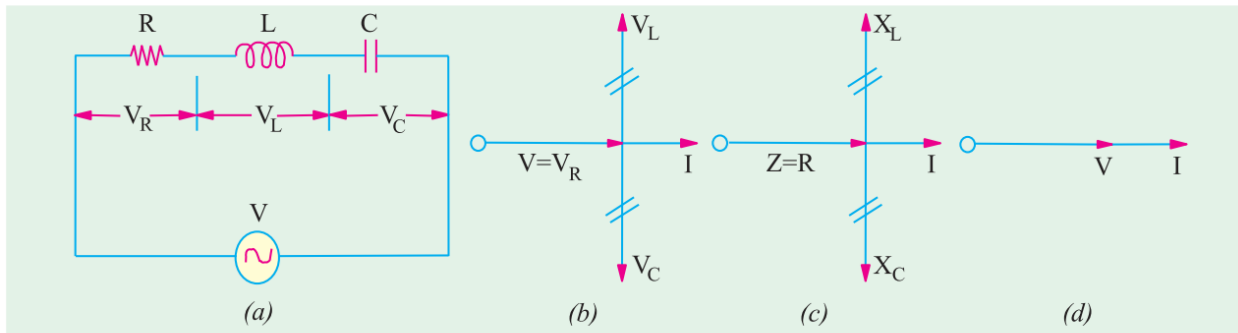


Fig. 13.44

When an R - L - C circuit is in resonance

1. net reactance of the circuit is zero *i.e.* $(X_L - X_C) = 0$. or $X = 0$.
2. circuit impedance is minimum *i.e.* $Z = R$. Consequently, circuit admittance is maximum.
3. circuit current is maximum and is given by $I_0 = V/Z_0 = V/R$.
4. power dissipated is maximum *i.e.* $P_0 = I_0^2 R = V^2/R$.
5. circuit power factor angle $\theta = 0$. Hence, power factor $\cos \theta = 1$.
6. although $V_L = V_C$ yet V_{coil} is greater than V_C because of its resistance.
7. at resonance, $\omega^2 LC = 1$
8. $Q = \tan \theta = \tan 0^\circ = 0^*$.